Akshat Ajay Banzal

cs24btech11005@iith.ac.in | akshatmbanzal@gmail.com | +91 9309884971 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.00
J.K. College of Science, Commerce and Arts, Pune	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education)	2024	92.33%
Pratibha International School , Chinchwad, Pune	Class X (Central Board of Secondary Education)	2022	99.00%

SCHOLASTIC ACHIEVEMENTS

- Cleared **IOQA** and the Indian National Astronomy Olympiad (**INAO**) to rank in the **top 55 of 13,000+** candidates nationwide, attending the Orientation Cum Selection Camp (**OCSC**) at **HBCSE Mumbai.**
- Recognized as **State Topper (Top 1%)** in the National Standard Examinations in Physics (**NSEP**) and Chemistry (**NSEC**).
- Qualified the Indian Olympiad Qualifier in Mathematics (**IOQM**).
- **Expert** on **Codeforces** (max rating - 1790)

WORK EXPERIENCE

FinStocks AI

Software Engineering Intern

- **Distributed Backend & Runtime-** Architected a **concurrent event-driven backend runtime** using Python (**asyncio**), PostgreSQL, **Redis Streams**, Pydantic, and **FSMs** to orchestrate **deterministic, fault-tolerant execution** of trading workflows.
- **Backtesting Engine** - Developed a **vectorized engine** using **partitioned Apache Parquet**, processing ~**2.9M** (4 months of 1-minute market data) time-series records in **<5s**.
- **Recommendation Engine** - Engineered scalable recommendation pipelines using LangChain/LangGraph, combining **quantitative screening** with LLM-driven analysis.

PROJECTS

Aurora Github

- Developed an **AI-powered pose correction pipeline** using **MediaPipe, SAM, OpenCV, Stable Diffusion, and ControlNet** enabling interactive human pose editing with **6-8 s local inference** on an **8 GB RAM smartphone**.
- Engineered the end-to-end pipeline involving **pose estimation, segmentation, keypoint transformation, geometric warping, and diffusion-based inpainting** and secured **5th place among 22 IITs** in the PS.

Slidely : Automated Presentation Generation Github

- Developed a **Canva extension** using **React**, **Go**, **Ollama (Gemma 4)**, and the **Canva App SDK** that converts raw text into formatted presentation slides on any user selected Canva template.
- Built a **Go middleware** for prompt construction and JSON schema validation.

Hardware-aware Hybrid Scheduler

- Built a **hardware-aware** hybrid Linux scheduler using **eBPF**, **PMU telemetry**, and **CPU affinity**.
- Reduced energy consumption by **11.7%** and average power draw by **27.1%** for **SPEC 2017** memory-bound workloads.
- Classified workloads using IPC, floating-point activity, and sleep/runtime ratio for **hybrid-core scheduling**.

Low-Latency C++ Order Book Github

- Built a sub-microsecond C++20 limit order book and matching engine for NASDAQ ITCH 5.0, sustaining **4.08M messages/s** with **159 ns p50, 333 ns p90, and 869 ns p99 latency.**
- Engineered a zero-allocation execution pipeline using a custom memory arena, intrusive linked lists, cache-line aligned data structures, and lazy deletion of price levels to achieve **O(1)** order allocation and cancellation.

Exploiting Floating-Point Units for Integer Acceleration

- Optimized integer workloads on **ARM Cortex-A76** using **ARM NEON** intrinsics, exploiting **SIMD** and **instruction-level parallelism** beyond compiler auto-vectorization.
- Achieved up to **2.68× speedup** over compiler **-O3** implementations on compute-bound workloads with **100% correctness**.
- Benchmarked **five algorithms**, performed **assembly-level analysis**, and identified **memory** and **control-flow bottlenecks**.

Campus Navigation Github

- Built a campus navigation web app using **React, Tailwind CSS, Node.js, Leaflet,** and **OpenStreetMap** for route planning.
- Developed an interactive map interface with real-time route generation between campus locations.

SKILLS

Languages: C++, C, Go, Python, SQL, RISC-V Assembly

Systems & Backend: Linux, Docker, Git, Redis Streams, PostgreSQL, Apache Parquet, asyncio, Pydantic

Frameworks & Libraries: React, Node.js, Express.js, LangChain, LangGraph, OpenCV, MediaPipe

RELEVANT COURSES

Computer Science: Data Structures & Algorithms, Computer Architecture, Operating Systems, Database Management Systems, Theory of Computation, Artificial Intelligence, Software Development Fundamentals.

Mathematics: Calculus, Probability & Random Variables, Linear Algebra, Transform Techniques, Statistics.

POSITIONS OF RESPONSIBILITY

Contingent Member for **Inter-IIT Tech Meet 14.0** held at **IIT Patna**.

Dev Core at **Lambda, The Software Development Club of IIT Hyderabad.**

Core at **Epoch, The AI-ML & Data Science Club of IIT Hyderabad.**

Web Coordinator at **FCC(Finance & Consulting Club) of IIT Hyderabad.**